

Police Complaint Documentation System

AI-Assisted Complaint Registration & Documentation

Project Review Presentation

Team Members: SHAMINI K - 24MID0263

MAGESH K - 24MID0253

ROHAN KARTHIK - 24MIC0123

RITHIK ROSHAN -24MID0300



Problem Statement & Field Interviews

The Problem

- Manual complaint documentation takes significant time
- Complaints given as natural, story-like descriptions
- Important information often missing or incomplete
- Writers must repeatedly ask clarifying questions

User Research Findings

> *"The system should assist documentation — not change the complainant's facts."*

Writer

Manual writing requires time; complainants lack structure; missing information requires follow-up

Inspector

Accuracy and verification are critical; final complaint must reflect complainant's actual statement; unsupported information must not be introduced

Core Problem & Evidence-Preserving Requirement

The Challenge

Complainants explain incidents naturally as stories.
Writers need structured information:

Evidence-Preserving Principles

✓ PRESERVE

Keep the original complaint statement intact

⬇ EXTRACT

Extract only information actually present in the statement

❓ ASK

Suggest questions when information is missing

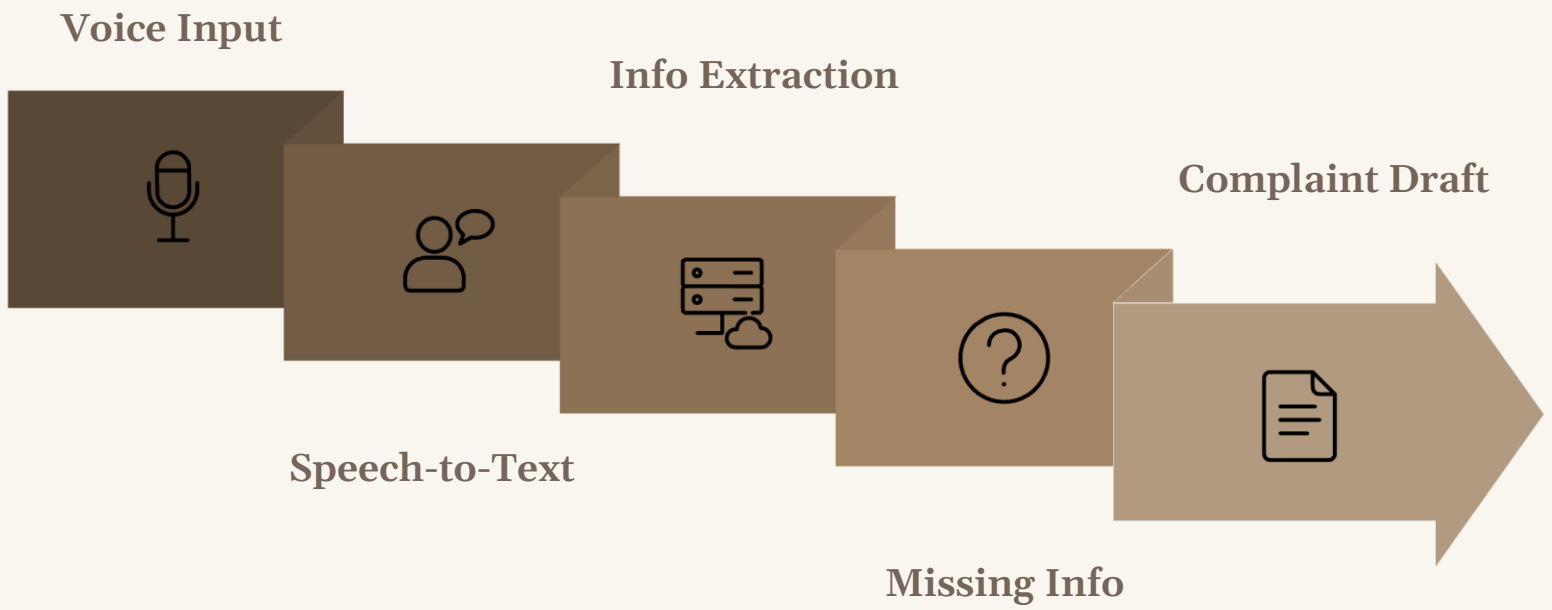
✗ NEVER ASSUME

Do not invent or guess missing information

Incident Type	**Object / Property**	**Date & Time**	**Location**
Person Involved		**Description**	

Proposed Solution & System Workflow

The system converts unstructured voice input into a structured, verified complaint draft while preserving the original statement.



MVP Scope & Features

✓ Included in MVP

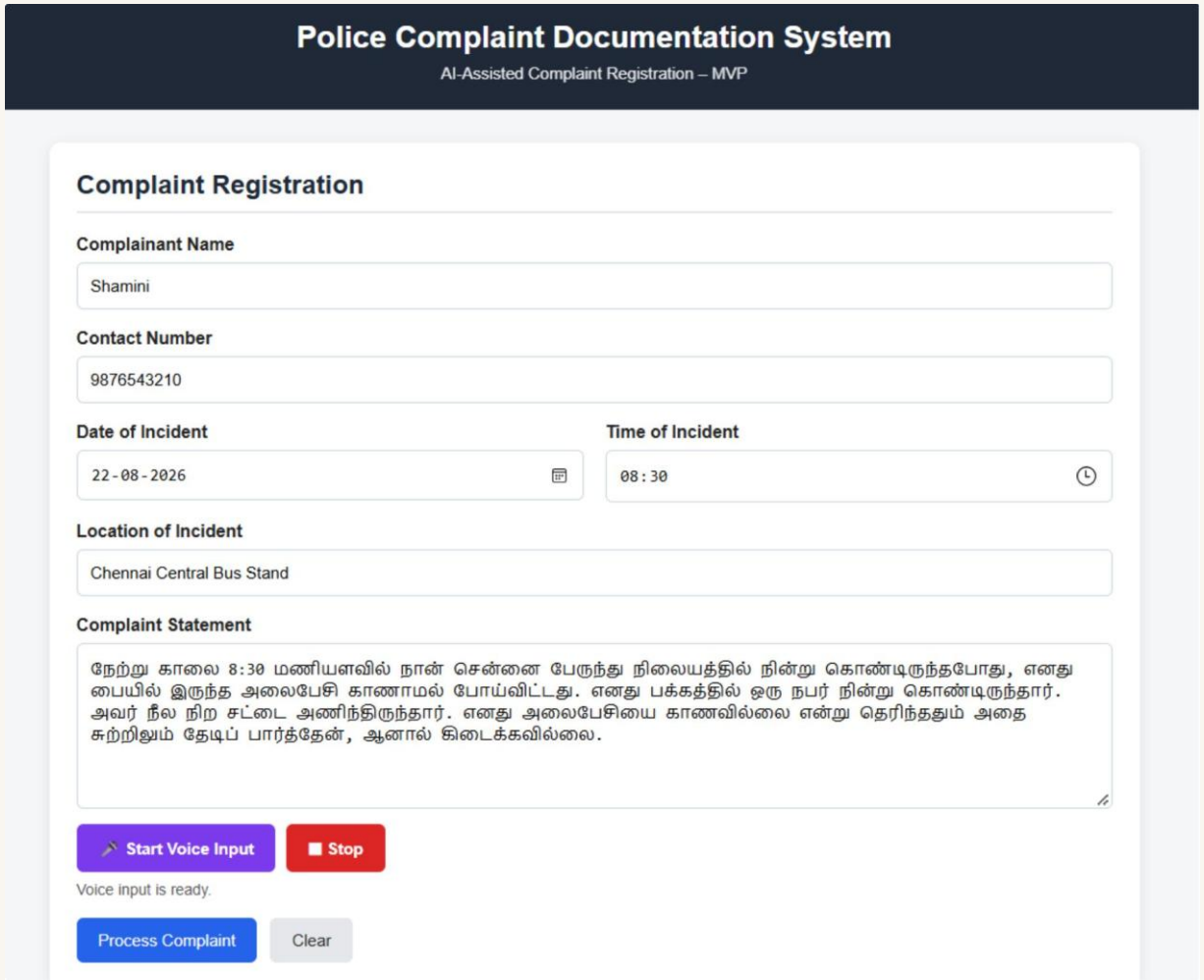
- Complaint details entry
- Voice input and Speech-to-text conversion
- Original complaint preservation
- Basic information extraction
- Missing-information detection & suggested questions
- Complaint / CSR draft generation
- Writer review and approval

→ Future Scope

- CCTNS Integration
- FIR Registration
- Police Database Integration
- Court Management
- Production-Level Deployment
- *And more...*

Working MVP Interface

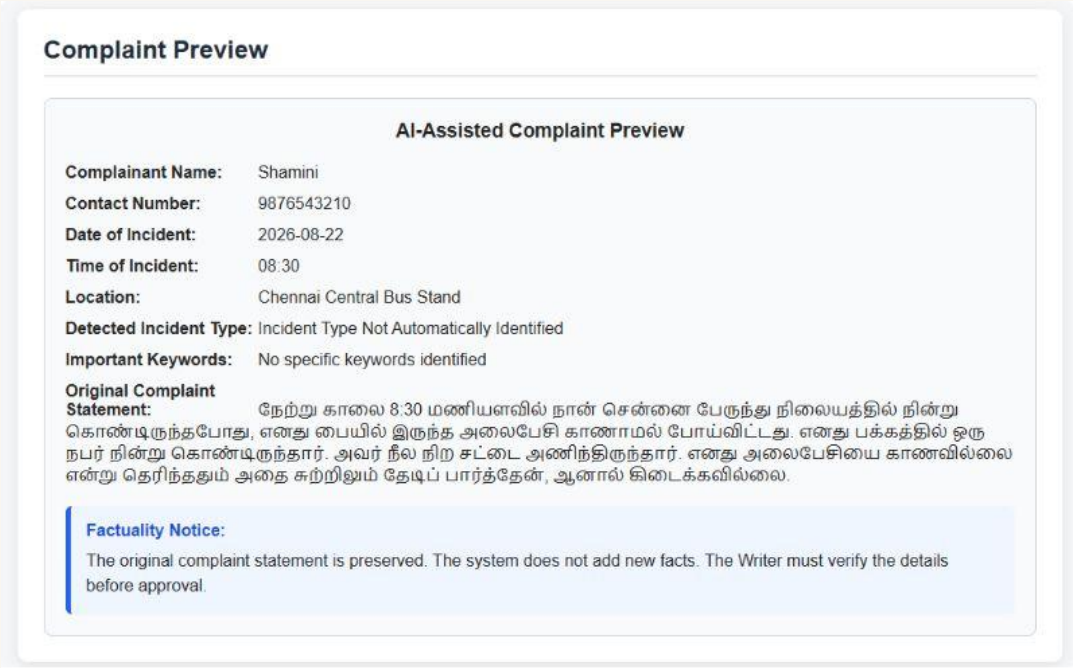
Complaint Registration Interface — Voice Input & Complaint Entry



MVP Demonstration: Natural Complaint to Structured Draft

System Processing Workflow: Input → Processing → Extraction → Missing Information Detection

Step 1: Complaint Preview & AI-Assisted Processing



Complaint Preview

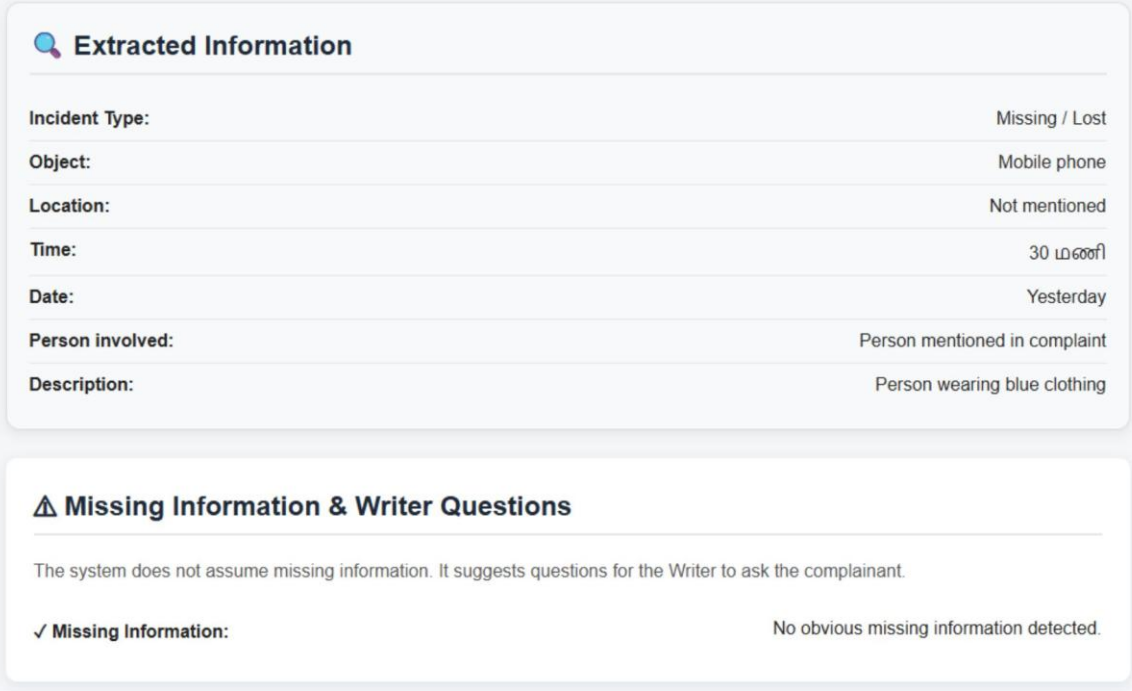
AI-Assisted Complaint Preview

Complainant Name:	Shamini
Contact Number:	9876543210
Date of Incident:	2026-08-22
Time of Incident:	08:30
Location:	Chennai Central Bus Stand
Detected Incident Type:	Incident Type Not Automatically Identified
Important Keywords:	No specific keywords identified

Original Complaint Statement: நேற்று காலை 8:30 மணியளவில் நான் சென்னை பேருந்து நிலையத்தில் நின்று கொண்டிருந்தபோது, எனது பையில் இருந்த அலைபேசி காணாமல் போய்விட்டது. எனது பக்கத்தில் ஒரு நபர் நின்று கொண்டிருந்தார். அவர் நீல நிற சட்டை அணிந்திருந்தார். எனது அலைபேசியை காணவில்லை என்று தெரிந்ததும் அதை சுற்றிலும் தேடிப் பார்த்தேன், ஆனால் கிடைக்கவில்லை.

Factuality Notice:
The original complaint statement is preserved. The system does not add new facts. The Writer must verify the details before approval.

Step 2: Information Extraction & Missing Information Detection



🔍 Extracted Information

Incident Type:	Missing / Lost
Object:	Mobile phone
Location:	Not mentioned
Time:	30 மணி
Date:	Yesterday
Person involved:	Person mentioned in complaint
Description:	Person wearing blue clothing

⚠ Missing Information & Writer Questions

The system does not assume missing information. It suggests questions for the Writer to ask the complainant.

✓ Missing Information:	No obvious missing information detected.

Experiment & Results

Test Cases

1. Missing / Lost Mobile Phone
2. Road Accident
3. Threat
4. Missing / Lost Bag
5. Missing / Lost Mobile Phone with Missing Details

Outcome

5 Test Cases → 5 Complaint Drafts Generated → 3 Usable Drafts

60%	70%
Achieved Rate	**Threshold**
3 of 5 drafts usable	Target not yet met

⚠ **Status:** Threshold NOT achieved — Improvements needed in information extraction accuracy and consistency

Generated Complaint Draft with Writer Verification

Writer Review & Approval

Review the complaint carefully before approval. The Writer is responsible for verifying the information.

Complaint Draft

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POLICE COMPLAINT / CSR DRAFT

BASIC INFORMATION
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Complainant Name: Shamini
Contact Number: 9876543210

INCIDENT INFORMATION
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Date of Incident: 2026-08-22
Time of Incident: 08:30
Location of Incident: Chennai Central Bus Stand
Incident Type: Missing / Lost
Object / Property: Mobile phone
Person Involved: Person mentioned in complaint
Description: Person wearing blue clothing

ORIGINAL COMPLAINT STATEMENT
-------------------------
நேற்று காலை 8:30 மணியளவில் நான் சென்னை பேருந்து நிலையத்தில் நின்று கொண்டிருந்தபோது, எனது
பையில் இருந்த அலைபேசி காணாமல் போய்விட்டது. எனது பக்கத்தில் ஒரு நபர் நின்று கொண்டிருந்தார்.
அவர் நீல நிற சட்டை அணிந்திருந்தார். எனது அலைபேசியை காணவில்லை என்று தெரிந்ததும் அதை
சுற்றிலும் தேடிப் பார்த்தேன், ஆனால் கிடைக்கவில்லை.

VERIFICATION NOTICE
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This draft is generated from information provided by the complainant.
No unsupported facts should be added.
The original complaint statement must be preserved.
Writer verification and approval are required.
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Limitations & Improvements Needed

What Needs Improvement

📍 Location Extraction

Inconsistent extraction of location information

📝 Description Extraction

Incomplete extraction of detailed descriptions

🔍 Incident Detection

Inconsistent incident type identification

🌐 Tamil Language

Natural Tamil expressions require better extraction rules

Next Focus Areas

- → Improve Tamil complaint understanding and NLP processing
- → Enhance date / time / location extraction accuracy
- → Strengthen entity and field extraction consistency
- → Improve evidence-preserving information processing

Project Progress & Future Direction

1

✓ COMPLETED

Problem Discovery

MVP Development

Initial Experiment

2

→ CURRENT

MVP Testing

Limitations Identified

60% Usable Rate

3

→ NEXT

NLP Improvement

Extraction Accuracy

Consistency Enhancement

4

→ FUTURE

Database Integration

Police Workflow

Advanced NLP

> *"The goal is to assist police documentation without changing the complainant's facts."*

Thank You

Questions & Discussion